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Business Groups and Tunneling: Evidence from Corporate Charitable Contributions by Korean Companies

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Abstract This paper investigates whether corporate philanthropic decisions are associated with a firm's listing status and business group affiliation. Analyzing a large sample of public and private firms in Korea, we find that (1) public firms make more charitable contributions than private firms and (2) business group-affiliated firms make more charitable contributions than non-affiliated firms. The results suggest that public firms, owing to greater public scrutiny, and business groups, owing to higher political costs, are encouraged to make more corporate charitable contributions. Further, we find that (3) greater corporate giving by public firms than private firms is more pronounced for business group-affiliated firms, compared with non-affiliated firms. The result is consistent with business groups' strategic coordination of their affiliates' philanthropic decisions to tunnel business group resources out to controlling shareholders who hold a larger portion of private affiliates than public affiliates.

Keywords Listing status · Business group · Corporate giving · Tunneling · Public scrutiny

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Introduction

Recently, interest in corporate social responsibility (CSR) has sharply increased in industry and academia. For example, the number of firms issuing CSR reports has dramatically increased (Dhaliwal et al. 2012) and numerous CSR studies have been published in accounting, finance, and management journals (e.g., Kim et al. 2012; Deng et al. 2013; Kang 2013). Among many dimensions of CSR, researchers pay increasing attention to corporate giving (i.e., charitable contributions) as a direct measure of CSR.[1] Prior research reveals various determinants of corporate giving: managers' altruism (Choi and Wang 2007; Masulis and Reza 2015), slack resources available to the firm (Seifert et al. 2004; Waddock and Graves 1997), firm size (Amato and Amato 2007), advertising intensity (Wang and Qian 2011), and even ownership structure (Atkinson and Galaskiewicz 1988; Barnea and Rubin 2010). While these studies shed some light on the motivations of corporate philanthropy, little is known about the impact of corporate governance on philanthropy. In particular, the literature has not addressed the influence of the listing status of firms on their corporate giving and has not examined the pattern of giving activities among business group members.

This study investigates the impact of listing status and business group affiliation on corporate philanthropic behavior. Korea provides a noble setting to investigate differences in corporate philanthropic behaviors between public and private firms, and those between business group-affiliated and non-affiliated firms. In Korea,

[1] Hillman and Keim (2001) argue that researchers need to exploit the individual components of CSR. Carroll (2004) depicts corporate philanthropy as one of the most important components of CSR.

companies report their charitable contributions on the audited financial statements. Thus, high-quality corporate giving data are available. Also, business groups, representing a collection of companies mutually affiliated under common control as if they are a single entity, are a popular type of business organization in Korea, as in other emerging economies (Khanna and Palepu 2000). This enables us to examine an interesting question of whether corporate giving by business group-affiliated firms differs from that of non-affiliated firms.

Using a sample of all public firms and a substantial body of private firms in Korea between 2000 and 2014, we first predict that public firms, owing to greater visibility and public scrutiny, are encouraged to actively engage in corporate philanthropy and find that public firms, on average, make greater charitable contributions than private firms. Next, we predict that business group-affiliated firms, owing to higher political costs, are more likely to engage in corporate philanthropy and find that group-affiliated firms make greater charitable contributions than non-affiliated firms. Lastly, we investigate strategic allocation of charitable contributions within the business group. Prior literature suggests that controlling shareholders of business groups exploit internal transactions among their affiliates for their benefit. In particular, Johnson et al. (2000) suggest that controlling shareholders expropriate minority shareholders' wealth to increase their own wealth and term such exploitation "tunneling". Bae et al. (2002) and Baek et al. (2006) show that controlling shareholders exploit financial and economic resources from mergers and acquisitions of, or private security offerings to other affiliated firms in the same business group. Extending these studies, we conjecture that controlling shareholders "tunnel" not only through financial transactions but also through corporate philanthropy. Consistent with our hypothesis, we find that public firms make a much larger amount of charitable contributions than private firms, especially when the firm belongs to a business group.

We conduct several additional analyses. First, we examine whether the control–ownership wedge, the disparity between control rights and cash flow rights, influences strategic allocation of corporate giving within the business group. We find that public firms of which control–ownership wedge is higher are likely to make a larger amount of charitable contributions, indicating that business groups strategically make use of charitable contributions as a channel of tunneling. Next, we examine whether a firm in a business group benefits from corporate giving by other affiliated firms in the same business group. To the extent that public affiliated firms make more charitable contributions than private affiliated firms, private affiliated firms are likely to benefit more from business group-wide giving than public affiliated firms. Consistent with our conjecture, we find that growth in public affiliated firms' charitable contributions is positively associated with subsequent revenue growth of private affiliated firms in the same business group. Finally, we examine temporal changes in the level of corporate giving when firms go public. We find that initial public offering firms increase their corporate giving until the public offering and gradually decrease their giving after that. But the levels of corporate giving in the post-public offering period still remain higher than those in the pre-public offering period, indicating that the listing status is an important determinant of corporate giving.

Our study contributes to the literature as follows. First, the study enriches our knowledge of the drivers of corporate philanthropy. While there are no prior studies, to our knowledge, that investigate the corporate giving behavior of private firms, we attempt to differentiate corporate philanthropic behaviors between public and private firms. Second, by examining corporate giving activities in Korea, where business groups are prevalent, we suggest a way to understand corporate philanthropic behavior in other emerging markets, where business groups are also a popular type of business organizations (Khanna and Palepu 2000). Finally, this study sheds light on the tunneling (business group) literature by showing that business groups strategically coordinate and allocate corporate philanthropy between public affiliates and private affiliates. In particular, we suggest an idea that controlling shareholders of business groups exploit group-wide corporate philanthropy as a non-financial channel of the tunneling process.

These findings also offer important practical implications. In particular, in an attempt to address wealth polarization problems, politicians have tabled a series of Chaebol reform policies including bans on cross-investments and cross-debt payment guarantees among business group's affiliates (e.g., Lee 2012). Moreover, the nation's civil society groups (see Kirk 2016) as well as foreign activist shareholders (see Frost 2007) closely monitor business groups and voice a variety of opinions in the name of the so-called 'minority shareholders' movement.' In response to the Chaebol-bashing sentiment and related policies, business groups have increased their charitable contributions and social engagements (see Choi 2010; Kim and Park 2012). Thus, our study gives investors, regulators, and policy makers insights into the motives of public firms and business groups to engage in extensive corporate giving.

The remainder of this paper is structured as follows. Section 2 discusses related research and develops the hypothesis. Section 3 describes our sample data and develops the research design. Section 4 presents our results and Sect. 5 presents additional analyses. Section 6 presents our conclusions.

Literature Review and Hypothesis Development

The Determinants of Corporate Philanthropic Behavior

Extant research suggests the following three explanations for corporate philanthropy: altruistic motivation; the principal-agent theory; and shareholder wealth maximization (Gautier and Pache 2015). First, as to the altruistic motivation, Haley (1991) suggests that corporate philanthropic behavior is mostly attributable to firms' altruism. Also, Choi and Wang (2007) argue that corporate philanthropy is the result of top managers' benevolence and integrity values.

Next, the agency theory suggests that firms use corporate giving opportunistically for their private benefits. Yermack (2009) documents that CEOs make charitable stock gifts to their family foundations just before share prices drop sharply in an attempt to increase their personal income tax benefits. Barnea and Rubin (2010) show that insider ownership is negatively related to a firm's CSR rating, suggesting that insiders (i.e., directors and blockholders) have an incentive to make a larger amount of charitable contributions when they own less.[2] Marquis and Lee (2013) suggest that short-tenured CEOs seek short-term benefits through philanthropy over long-term firm objectives. Recently, Masulis and Reza (2015) report a negative association between a CEO's fractional ownership and corporate giving, supporting the agency motives of corporate philanthropy. Further, research on the rent-seeking behavior of owner-managers suggests that controlling shareholders enjoy benefits of charitable giving at the expense of minority shareholders (Li et al. 2015b; Tan and Tang 2014).

Finally, the shareholder wealth maximization perspective suggests that corporate philanthropy helps to improve corporate financial performance. According to Margolis and Walsh (2003), among 109 studies that examined the relationship between the level of corporate social performance and financial performance, 54 studies report a positive relationship, implying that corporate social activities, on average, enhance future firm performance.[3] More directly, Lev et al. (2010) document that growth in corporate philanthropy is followed by future revenue growth. Another stream of studies supports the shareholder wealth maximization perspective by documenting that managers use corporate philanthropy as a means of corporate marketing. In particular, Schwartz (1968) and Wang and Qian (2011) show that corporate philanthropy is a substitute for corporate advertisement, supporting the idea that corporate giving leads to higher firm values. Additionally, broadening the shareholder wealth maximization perspective to that of stakeholders, Aguilera et al. (2007) argue that corporate social activity is driven by a variety of stakeholders' economic and non-economic motives. Several empirical studies (e.g., Campbell and Slack 2006; Gan 2006; Brammer and Millington 2005; Yoo and Pae 2016) support this notion by showing that firms with high public scrutiny make more charitable contributions than firms with low public scrutiny.

The Effect of Listing Status and Group Affiliation on Corporate Philanthropic Behavior

Researchers have documented differences in corporate decisions between public and private firms. For example, Ball and Shivakumar (2005), Hope et al. (2013), and Kim and Yi (2006) investigate the effect of listing status on financial reporting quality in the U.K., U.S. and Korea. Also, Chaney et al. (2004) examine the audit fee premium on Big N auditors and find that the so-called Big N premium does not exist among private firms. Overall, extant studies comparing public and private firms argue that different demands of investors and creditors account for different corporate decisions. Private firms without external equity investors are under less heavy public scrutiny than public firms and thus have weaker incentives to manage earnings and pay an audit fee premium.

Extant research suggests a link between a firm's listing status and its visibility. Theoretically, Merton (1987) suggests that firm visibility (i.e., investor recognition) is linked with its listing status because public firms tend to have a wider investor base than private firms. Empirically, Kadlec and McConnell (1994) and Foerster and Karolyi (1999) test Merton's investor recognition hypothesis and show that investor recognition and interest (i.e., visibility) increase when firms go public. Mehran and Peristiani (2010) examine public firms' decision to go private and show that initial public offering (IPO) firms are more likely to go private (i.e., delist) when they fail to sustain investor recognition or visibility. A survey by Bancel and Mittoo (2009) reports that European chief financial officers (CFO) view enhanced visibility and investor recognition as the most important benefit of going public.

Some studies investigate the effect of public scrutiny on corporate philanthropy (Gan 2006; Campbell and Slack 2006). Using court cases and news articles as proxies for public scrutiny, Gan (2006) shows that firms' vulnerability to public scrutiny drives corporate philanthropy. Campbell and Slack (2006) suggest that highly visible firms are likely

[2] Note that Barnea and Rubin (2010) do not directly examine charitable contributions. Rather, they examine CSR ratings assuming that the rating is associated with CSR-related expenditure level.

[3] Of the remaining 55 papers, 7 studies found a negative relationship, 28 studies brought insignificant associations, and 20 studies reported mixed results.

to make more corporate charitable contributions. Also, Brammer and Millington (2005) suggest that firms make charitable contributions to gain corporate reputation. Overall, extant studies suggest that highly visible firms have strong incentives to make more charitable contributions.

While prior studies have examined various incentives for corporate philanthropy of public firms, little has been known about those of private firms despite their economic importance. Considering public firms are under greater public scrutiny (Ball and Shivakumar 2005; Kim and Yi 2006) and public visibility and scrutiny induce a higher level of corporate philanthropy (Gan 2006; Campbell and Slack 2006), we expect that public firms make more charitable contributions than private firms. Our first hypothesis concerns the effect of listing status on corporate philanthropy:

H1 Public firms make more charitable contributions than private firms.

After the Asian financial crisis in 1997, the Korean government amended business group-related regulations. The revised regulations limit mutual investments among group-affiliated firms to mitigate the concentration of controlling shareholders' voting rights. Further, the Korean Fair Trade Commission (KFTC) curbs loan guarantees among group-affiliated firms to prevent those firms from becoming insolvent at the same time.[4] However, the Korean government is still criticized for favoring business groups with lax regulations and various types of financial subsidies. For example, in the telecommunication industry, the government granted the right to use indispensable radio frequencies only to those firms affiliated with large business groups.

Watts and Zimmerman (1978) argue that political costs, which include all expected costs imposed on a firm from potential adverse political actions (e.g., antitrust, regulation, government subsidies, taxes, and tariffs), affect various corporate decisions. Business groups in Korea have been suspected of announcing big donations and setting up a charity foundation in an attempt to secure the government's favor rendering, such as relaxed regulations and financial subsidies. Further, business groups have been criticized as the main culprit behind the structural weakness of the Korean economy, which had led to the nation's financial crisis in 1997 (Moskalev and Park 2010; Campbell II and Keys 2002).

Taken together, we expect that group-affiliated firms make more charitable contributions than non-affiliated firms in order to maintain a good rapport with the government and society. Our second hypothesis concerns the effect of business group affiliation on corporate philanthropy.

H2 Business group-affiliated firms make more charitable contributions than non-affiliated firms.

Internal Business Transactions (i.e., Tunneling) and Philanthropy

Business groups make coordinated decisions to maximize the value of the group as a whole. If public firms, which are under greater public scrutiny (H1), and business groups, which are subject to higher political costs and visibility costs (H2), make more charitable contributions, then it is likely that business groups coordinate their corporate philanthropy strategy such that their public affiliates make more charitable contributions than private affiliates. In this section, we argue that incentives for greater corporate philanthropy by public firms than private firms are more pronounced for business group-affiliated firms than non-affiliated firms because business groups may coordinate their group-wide philanthropy strategy for the benefits of controlling shareholders of the business group.

Business groups are prone to tunneling since business groups are often controlled by a single shareholder or a group of closely related parties. Prior literature suggests that controlling shareholders of business groups often strategically assign resources within business groups to themselves via internal business transactions. For instance, in Korea, Bae et al. (2002) provide evidence of systematic tunneling within business groups in mergers and acquisitions; Baek et al. (2006) provide the evidence of tunneling in private security offerings. Similarly, in India, Bertrand et al. (2002) find that controlling shareholders tunnel profits from affiliated firms for which they have low cash flow rights to the firms for which they have high cash flow rights.

As discussed above, the extant literature on tunneling mainly focuses on managers' expropriation of financial resources. However, Chang and Hong (2000) show that business groups strategically allocate not only financial resources but also intangible resources such as technology and brand loyalty among affiliates. Further, Wang and Qian (2011) show that firms utilize corporate philanthropy as a substitute for marketing practices. We contend that business groups will engage in corporate philanthropy mainly through their public affiliates rather than private affiliates because society expects greater corporate social responsibility from public firms; moreover, corporate giving through public firms is a more effective way to enhance the group's public image and reputation as a socially responsible business entity. Taken together, we predict that controlling shareholders of business groups will strategically

[4] Details are available at KFTC's business group information portal (http://groupopni.ftc.go.kr/).

allocate a larger amount of corporate charitable contributions to public affiliates than private affiliates. That is, our finding in H1 will be more pronounced within a business group which has both public and private firms as its affiliates. Our third hypothesis concerns the difference of corporate charitable contributions between public and private affiliated firms within business groups:

H3 The influence of listing status on corporate philanthropy (i.e., H1 or the larger charitable contribution by public firms than private firms) is more clearly manifested for business group-affiliated firms than for non-affiliated firms.

Research Design

Sample Selection and Data

Our initial sample consists of all public and private firms that are included in the Korea Listed Companies Association (KLCA) database. KLCA provides profiles and financial information including corporate charitable contributions reported in the financial statements for all public firms and a substantial body of private firms in Korea. Initially, we collect 162,430 firm-year observations over the period from 2000 to 2014. We exclude observations with insufficient data on such firm characteristics as total assets, liabilities, equities, sales, net profit, cash flow, and the age of the firm. Further, we delete observations without data for charitable contributions (e.g., Chen et al. 2008; Li et al. 2015a, 2015b). These screening procedures yield a final sample of 69,558 observations.

We classify a firm as a public firm if its shares are listed on the Korea Exchange (KRX). Private firms are those not listed on KRX, but subject to a mandatory external audit. The *Act on External Auditing in Korea* requires all firms with total assets of seven billion Korean Won (KRW) or above to have their financial statements audited by independent auditors.[5] Thus, all public and private firms in our sample are subject to the same reporting and external audit requirements, and there is no difference in the corporate tax rates between public and private firms.

Following previous literature (e.g., Baek et al. 2006), we identify each firm's group affiliation using the KLCA database. We classify a firm as an affiliate if it belongs to a business group which has at least five member firms. Further, we conduct robustness tests, varying proxies for business groups in Sect. 5.5.[6]

Regression Model

Factors that affect a firm's decision to go public may also influence its corporate philanthropy. That is, the endogenous selection of going public or remaining private could affect the association between listing status and charitable contributions. To address this potential self-selection bias, we employ Heckman's (1979) two-stage procedure following the prior literature.[7]

In the first stage, we estimate a Probit model for a firm's choice of listing status. Following Ball and Shivakumar (2005) and Kim and Yi (2006), we include the following explanatory variables: *FirmSize*, *Leverage*, *ExportRatio*, *QuickRatio*, *SalesGrowth*, and *Age*. Chemmanur and Fulghieri (1999) report that young and small firms facing severe information asymmetry are less likely to choose to go public, implying that *FirmSize* (the natural log of total assets) and *Age* (the number of years since the firm is established) will be positively associated with listing status.

Pagano et al. (1998) and Albornoz and Pope (2004) show that investors are reluctant to invest in financially distressed companies, implying that *Leverage* (total liabilities divided by the book value of equity) and *QuickRatio* (current assets less inventory divided by current liabilities) will be associated with a firm's choice of listing status. Chemmanur et al. (2010) report that growing firms are more likely to go public, implying that *SalesGrowth* (Percentage change in sales) is positively associated with listing status. We also include *ExportRatio* (the ratio of the amount of exports to total sales) because companies with greater proportion of foreign business are more likely to go public in Korea.

Further, we include two additional instrument variables which are associated with the likelihood of listing, but less likely to be associated with corporate giving: the degree of market competition (the *Herfindahl–Hirschman Index* or *HHI*) and the proportion of public firms in the industry (*Industry_PubPr*). Chemmanur and He (2011) suggest that firms are more likely to go public if the level of market competition is higher, that is, *HHI* is lower.[8] Also, the proportion of public firms in an industry is likely to be associated with the likelihood of the listing status of a firm in the industry. We measure these variables as of year $t-1$, since a firm's listing choice in year t is based on the

[5] The total asset requirement of KRW 7 billion has changed to KRW 10 billion in 2009, and KRW 12 billion in 2015.

[6] In untabulated tests, we verify that our results remain qualitatively similar when we change the minimum number of affiliated firm in a business group from five to two or to ten.

[7] Prior literature uses Heckman procedure instead of the often used two-stage least squares (2SLS) method when the variable of interest is not a continuous variable, but a binary variable (e.g., Ball and Shivakumar 2005; Kim and Yi 2006; Givoly et al. 2010). Wooldridge (2010) shows that non-OLS regression models as the first-stage equation of 2SLS estimation do not provide consistent coefficient estimates.

[8] The *Herfindahl–Hirschman Index* (HHI) is a measure of the market concentration of an industry. Thus, highly competitive industries have a relatively low HHI.

financial characteristics in year $t-1$. In particular, we run the following first-stage Probit regression:

$$\begin{aligned} Pr\left(Public_{i,t}=1\right) = \Phi(\alpha_0 &+ \alpha_1 FirmSize_{i,t-1} \\ &+ \alpha_2 Leverage_{i,t-1} + \alpha_3 ExportRatio_{i,t-1} \\ &+ \alpha_4 QuickRatio_{i,t-1} + \alpha_5 SalesGrowth_{i,t-1} + \alpha_6 Age_{i,t-1} \\ &+ \alpha_7 Industry\_PubPr_{i,t-1} + \alpha_8 HHI_{i,t-1} + \varepsilon) \end{aligned} \quad (1)$$

In the second stage, we estimate the following OLS regression model including the *inverse Mills ratio* from the first-stage regression to examine the influence of listing status and business group affiliation on corporate charitable contributions:

$$\begin{aligned} Giving_{i,t} = \beta_0 &+ \beta_1 Public_{i,t} + \beta_2 FirmSize_{i,t} + \beta_3 Age_{i,t} \\ &+ \beta_4 Leverage_{i,t} + \beta_5 ROA_{i,t} + \beta_6 Advertising_{i,t} \\ &+ \beta_7 PreGivingD_{i,t} + \beta_8 Big4Audit_{i,t} \\ &+ \beta_9 IMR + (\text{Year dummies}) \\ &+ (\text{Industry dummies}) + \varepsilon \end{aligned} \quad (2)$$

$$\begin{aligned} Giving_{i,t} = \gamma_0 &+ \gamma_1 Affiliate_{i,t} + \gamma_2 FirmSize_{i,t} + \gamma_3 Age_{i,t} \\ &+ \gamma_4 Leverage_{i,t} + \gamma_5 ROA_{i,t} + \gamma_6 Advertising_{i,t} \\ &+ \gamma_7 PreGivingD_{i,t} + \gamma_8 Big4Audit_{i,t} \\ &+ \beta_9 IMR + (\text{Year dummies}) \\ &+ (\text{Industry dummies}) + \varepsilon \end{aligned} \quad (3)$$

$$\begin{aligned} Giving_{i,t} = \delta_0 &+ \delta_1 Public_{i,t} + \delta_2 Affiliate_{i,t} + \delta_3 Public_{i,t} \\ &\times Affiliate_{i,t} + \delta_4 FirmSize_{i,t} + \delta_5 Age_{i,t} \\ &+ \delta_6 Leverage_{i,t} + \delta_7 ROA_{i,t} \\ &+ \delta_8 Adverti\sin g_{i,t} + \delta_9 \Pr eGivingD_{i,t} \\ &+ \delta_{10} Big4Audit_{i,t} + \delta_{11} IMR + (\text{Year dummies}) \\ &+ (\text{Industry dummies}) + \varepsilon, \end{aligned} \quad (4)$$

where $Giving_{i,t}$ is the amount of charitable contribution as a percentage of sales; $Public_{i,t}$ is an indicator variable that equals one if a firm is listed; and $Affiliate_{i,t}$ is an indicator variable that equals one if a firm belongs to a business group.

Our main variables of interest are *Public, Affiliate,* and the interaction term between the two variables (i.e., *Public* × *Affiliate)*. The first (second) hypothesis examines whether public (group-affiliated) firms make a larger amount of charitable contributions than private (or non-affiliated) firms. Accordingly, we predict that the coefficient estimates on *Public* in Eq. (2) and on *Affiliate* in Eq. (3) will be positive (i.e., H1: $\beta_1 > 0$ and H2: $\gamma_1 > 0$). The third hypothesis examines whether business groups strategically allocate charitable contributions more to public affiliates than private affiliates. That is, we compare the amount of charitable contributions by public firms in excess of that of private firms between business group affiliates and non-affiliates. We predict that the coefficient on the interaction term in Eq. (4) will be positive (i.e., H3: $\delta_3 > 0$) (see Appendix 2 for detailed discussion of the statistical model).[9]

In examining the influence of listing status and business groups on corporate charitable contributions, we control for several factors such as *FirmSize, Age, Leverage, ROA, Advertising, Big4Audit* and *PreGivingD. FirmSize* (the natural log of total assets) is the most important institutional determinant of charitable contributions (Chen et al. 2008; Seifert et al. 2003). *Age* (the number of years since the firm is established) is also an important determinant of charitable contributions (Roberts 1992; Moore 2001). Brown et al. (2006) show that monitoring by creditors curbs managers' rent extraction behaviors, implying that *Leverage* (total liabilities divided by the book value of equity) is negatively associated with the amount of charitable contributions. Waddock and Graves (1997) suggest that profitable firms are likely to make more charitable contributions, indicating that *ROA* (net income divided by total assets) is positively associated with the amount of charitable contributions. Prior research reports that firms make more charitable contributions when they spend a large amount on advertisement and product promotion (Fry et al. 1982; Brown et al. 2006; Zhang et al. 2010). To control for the quality of external monitoring, we include *Big4Audit* (an indicator variable that equals one if a firm has a Big 4 auditor). Last, we include *PreGivingD* (an indicator variable that equals one if a firm made charitable contributions in the preceding year) to control for recurring characteristics of corporate giving. Prior studies show that there is a cross-sectional variation in charitable contributions across industries (Amato and Amato 2007; Zhang et al. 2010). Accordingly, we include industry dummy variables (classified by two-digit industry classification codes). Further, to control for potential year effects, we add year dummy variables. We also note that *Giving* is highly skewed. To mitigate the impact of the skewed distribution of *Giving*, we restrict our analyses to observations whose corporate giving is within three standard deviations of the mean value of corporate giving.[10] All continuous variables are winsorized at the 1 and 99% level to reduce the effects of outliers.

Furthermore, we ensure the robustness of our tests by conducting two alternative matching techniques: a propensity-score matching (Rosenbaum and Robin 1983) and a coarsened exact matching (Iacus et al. 2011).

Propensity-score matching (PSM) selects the matching private firms based on the probability estimate of being public

[9] Our prediction based on H3 ($\delta_3 > 0$) is also consistent with greater charitable contributions by group-affiliated firms than non-affiliated firms being more pronounced for public firms than for private firms.

[10] Our results are qualitatively similar when we add the excluded observations back to the sample.

Table 1 Descriptive statistics

Variable	Mean	STD	$Q1$	Median	$Q3$
Panel A: Total sample ($N = 69{,}558$)					
$Giving_t$	0.077	0.127	0.006	0.023	0.083
$Public_t$	0.194	0.395	0	0	0
$Affiliate_t$	0.071	0.257	0	0	0
$FirmSize_t$	3.739	1.262	2.786	3.432	4.417
Age_t	20.694	12.569	11	18	28
$Leverage_t$	2.405	3.993	0.565	1.248	2.531
ROA_t	0.059	0.082	0.019	0.051	0.095
$Advertising_t$	0.006	0.016	0	0	0.003
$PreGivingD_t$	0.863	0.344	1	1	1
$Big4Audit_t$	0.216	0.411	0	0	0

Variable	Public ($N = 13{,}476$)		Private ($N = 56{,}082$)		p value for the tests of the difference between the two groups	
	Mean	Median	Mean	Median	t test	Wilcoxon Z test
Panel B1: Sub-samples: public versus private						
$Giving_t$	0.086	0.028	0.075	0.022	<0.01	<0.01
$Affiliate_t$	0.161	0	0.050	0	<0.01	<0.01
$FirmSize_t$	5.023	4.809	3.431	3.185	<0.01	<0.01
Age_t	28.892	28	18.724	16	<0.01	<0.01
$Leverage_t$	1.100	0.735	2.719	1.462	<0.01	<0.01
ROA_t	0.045	0.044	0.063	0.053	<0.01	<0.01
$Advertising_t$	0.007	0.001	0.005	0	<0.01	<0.01
$PreGivingD_t$	0.928	1	0.847	1	<0.01	<0.01
$Big4Audit_t$	0.516	1	0.144	0	<0.01	<0.01

Variable	Affiliates ($N = 4955$)		Non-affiliates ($N = 64{,}603$)		p value for the tests of the difference between the two groups	
	Mean	Median	Mean	Median	t test	Wilcoxon Z test
Panel B2: Sub-samples: Business group affiliates versus non-affiliates						
$Giving_t$	0.101	0.037	0.075	0.022	<0.01	<0.01
$Public_t$	0.439	0	0.175	0	<0.01	<0.01
$FirmSize_t$	5.590	5.580	3.597	3.349	<0.01	<0.01
Age_t	23.111	20	20.509	18	<0.01	<0.01
$Leverage_t$	1.932	1.115	2.441	1.260	<0.01	<0.01
ROA_t	0.060	0.052	0.059	0.051	0.60	0.44
$Advertising_t$	0.004	0	0.006	0	<0.01	0.07
$PreGivingD_t$	0.930	1	0.857	1	<0.01	<0.01
$Big4Audit_t$	0.626	1	0.184	0	<0.01	<0.01

Variable = Giving	Listing status				p value for the tests of the difference between the two groups	
	Public		Private			
	Mean	Median	Mean	Median	t test	Wilcoxon Z test
Panel C: Comparison of *Giving* by business group affiliation and listing status (2×2) ($N = 69{,}558$)						
Business group affiliation						
Affiliates	0.105 ($N = 2173$)	0.044	0.098 ($N = 2782$)	0.033	0.08	<0.01
Non-affiliates	0.082 ($N = 11{,}303$)	0.026	0.074 ($N = 53{,}300$)	0.021	<0.01	<0.01

Table 1 continued

Variable = Giving	Listing status				*p* value for the tests of the difference between the two groups	
	Public		Private			
	Mean	Median	Mean	Median	*t* test	Wilcoxon *Z* test
p value for the tests of the difference between the two groups	<0.01	<0.01	<0.01	<0.01		

This table documents descriptive statistics for our sample during the period 2000–2014. Panel A presents the summary statistics for the full sample. Panel B1 and B2 report the comparison for variables used in our analyses between public and private firms, and business group affiliates and non-affiliates, respectively. Panel C provides the comparison of the means and medians of corporate giving by business group affiliation and listing status. All continuous variables are winsorized at the 1% and 99% level

***, **, and * denote statistical significance at the 1, 5, and 10% level, respectively. See Appendix 1 and Sect. 3.2 for variable definitions

Table 2 Pearson correlation matrix

		[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]
(N = 69,558)										
$Giving_t$	[1]	0.034***	0.053***	0.063***	0.047***	−0.030***	0.053***	0.109***	0.105***	0.034***
$Public_t$	[2]		0.170***	0.499***	0.320***	−0.160***	−0.087***	0.044***	0.092***	0.358***
$Affiliate_t$	[3]			0.406***	0.053***	−0.033***	0.002	0.017***	0.055***	0.276***
$FirmSize_t$	[4]				0.332***	−0.084***	−0.000	0.067***	0.158***	0.425***
Age_t	[5]					−0.148***	−0.116***	−0.029***	0.132***	0.138***
$Leverage_t$	[6]						−0.194***	−0.031***	−0.066***	−0.084***
ROA_t	[7]							0.102***	0.013***	−0.012***
$Advertising_t$	[8]								0.037***	0.053***
$PreGivingD_t$	[9]									0.076***
$Big4Audit_t$	[10]									

This table presents the Pearson correlation matrix for the variables used in our analyses. All continuous variables are winsorized at the 1% and 99% level

***, **, and * denote statistical significance at the 1, 5, and 10% level, respectively. See Appendix 1 and Sect. 3.2 for variable definitions

rather than private using a propensity-score model. As instrumental variables of the propensity-score model, we employ variables used in the first and second stages of the Heckman model, *i.e.*, *FirmSize*, *Leverage*, *ExportRatio*, *QuickRatio*, *SalesGrowth*, *Age*, *ROA*, *Advertising*, *Big4Audit* and *PreGivingD*. PSM yields a balanced sample of 16,000 (8000 public and 8000 private) firm-year observations.

Coarsened exact matching (CEM) selects the matching private firms based on the coarsened strata of the covariates.[11] We coarsen each of the continuous covariates into eight equal-sized intervals. CEM yields a sample of 39,054 (7292 public and 31,762 private) firm-year observations.

[11] Recently, researchers have raised concerns and criticism over the misuse of the propensity-score matching in addressing endogeneity issues (see DeFond et al. 2016; Shipman et al. 2016). That is, seemingly innocuous changes in PSM design can significantly affect sample composition and inferences from the results. DeFond et al. (2016) suggest using CEM as an alternative matching approach to PSM.

Empirical Results

Descriptive Statistics

Panel A of Table 1 reports descriptive statistics. An average sample firm makes charitable contributions of KRW 770 per sales of KRW 1 million. Note that the median of *Giving* (0.023) is less than a third of the mean value (0.077), suggesting that the distribution of *Giving* is right skewed. Public and affiliated firms account for 19.4 and 7.1% of the sample, respectively. The average debt-to-equity ratio (*i.e.*, *Leverage*) is 2.4, while the average age of firms is 20.7 years.

In Panel B1, we partition the sample into public firms and private firms. We have 13,476 firm-year observations for public firms and 56,082 firm-year observations for private firms. Public firms make more charitable contributions than private firms: the mean (median) *Giving* is 0.086 (0.028) for public firms and 0.075 (0.022) for private firms. Public firms also differ from private firms in other dimensions. Public firms are significantly larger than private firms: the mean *FirmSize* is 5.023 for public firms and 3.431 for private firms. Public firms are more likely to be a member of a business group than private firms: 16.1% of public firms are affiliated with business groups, whereas 5.0% of private firms are affiliated.

In Panel B2, we partition the sample into group-affiliated and non-affiliated firms. We have 4955 firm-year observations for affiliated firms and 64,603 firm-year observations for non-affiliated firms. Affiliated firms make more charitable contributions than non-affiliated firms: the mean (median) *Giving* is 0.101 (0.037) for affiliated firms and 0.075 (0.022) for non-affiliated firms. Affiliated firms also differ from non-affiliated firms in other dimensions. Affiliated firms are significantly larger than non-affiliated firms: the mean *FirmSize* is 5.590 for affiliated firms and 3.597 for non-affiliated firms. Affiliated firms are more likely to be public firms: 43.9% of affiliated firms are listed, whereas 17.5% of non-affiliated firms are listed. However, the mean *ROA* of affiliated firms is not significantly different from that of non-affiliated firms, suggesting that business group-affiliated firms are not more profitable than non-affiliated firms.

Panel C provides the comparison of *Giving,* conditional on listing status and group affiliation. We partition the sample firms into four sub-samples, based on whether they are listed and whether they are affiliated with business groups. We find that the mean and median values of *Giving* in the *Public* and *Affiliate* sub-sample are significantly higher than those in the other three sub-samples (i.e., the *Private* and *Affiliate* sub-sample, the *Public* and *Non-affiliate* sub-sample, and the *Private* and *Non-affiliate* sub-sample). Further, among business group affiliates, public firms make more charitable contributions than private firms: the mean (median) *Giving* of 0.105 (0.044) for public firms versus that of 0.098 (0.033) for private firms. A similar pattern is observed for non-affiliates. We also note that the difference in the median values of *Giving* between the public and private firms for business group affiliates (0.011 = 0.044 − 0.033) is greater than that for non-affiliates (0.005 = 0.026 − 0.021). The result is consistent with our third hypothesis that business groups strategically allocate charitable contributions more to public affiliates than to private affiliates.

Table 2 reports the Pearson correlations. Consistent with evidence in Panel B1 (B2) of Table 1, *Public* (*Affiliate*) is significantly positively associated with *Giving,* supporting our first (second) hypothesis that public (affiliated) firms make a larger amount of charitable contributions than private (non-affiliated) firms. Further, *Giving* shows its strong correlations with various firm-specific characteristics, suggesting that it is important to control for firm characteristics in the multivariate regression analysis.

Multivariate Analysis

Table 3 presents the regression results for the influence of listing status and group affiliation on corporate charitable contributions. Panel A reports the second-stage regression results of the Heckman approach for the influence of listing status and group affiliation on corporate charitable contributions.[12,13] We use *Giving*, the amount of charitable contribution expressed in the percentage of sales, as the dependent variable. Model (1) includes *Public* to examine the impact of listing status on charitable contributions. Model (2) includes *Affiliate* to investigate the impact of group affiliation on charitable contributions. Model (3) include *Public, Affiliate,* and the interaction term of the two variables to examine whether business groups allocate a larger amount of charitable contributions to their public affiliates.

In model (1), we find that the coefficient on *Public* (β_1) is positive and statistically significant at the 0.05 level, supporting H1: that public firms make a larger amount of charitable contributions than private firms. In model (2), the coefficient on *Affiliate* (γ_1) is also positive and statistically significant at the 0.01 level, supporting H2: that business group affiliates make a larger amount of charitable contributions than non-affiliates. In model (3), the coefficient on the interaction term of *Public* and *Affiliate* (δ_3) is positive and statistically significant at the 0.05 level. The result supports H3: that higher levels of charitable giving by public firms vis-à-vis private firms are more pronounced among group-affiliated firms than non-affiliates. It is also consistent with higher levels of charitable giving by group-affiliated firms vis-à-vis non-affiliated firms being more pronounced among public firms than private firms.

Further, in model (3), we find that the coefficient on *Public* (δ_1) is statistically insignificant, suggesting that the amount of charitable contributions is not significantly different between public and private firms when they are not affiliated with business groups. But, among business group affiliates, public firms do make more charitable contributions than private firms: the value of $\delta_1 + \delta_3$ (0.011) is significant at the 0.01 level. Taken together, these results indicate that the significant coefficient on *Public* (β_1) in model (1) is mainly driven by the charitable giving behavior of business group-affiliated public firms.

[12] In the first-stage estimates of Heckman approach, we find that the coefficients on *FirmSize, Leverage, ExportRatio, QuickRatio, Age, Industry_PubPr, and HHI* (except *SalesGrowth*) are highly significant, indicating that our prediction model for a firm's listing status is well specified.

[13] To address concern about the heteroscedasticity of error terms, we also apply the White adjustment for standard errors, but the results are qualitatively similar to what we present in the paper.

Table 3 Influence of listing status and group affiliation on charitable contribution

Dependent variable = $Giving_t$	Model (1)		Model (2)		Model (3)	
	Coeff.	*p* value	Coeff.	*p* value	Coeff.	*p* value
Panel A: OLS regression of the amount of corporate giving on listing status and group affiliation results using Heckman approach ($N = 69{,}558$)						
Intercept	0.066***	0.00	0.070***	0.00	0.070***	0.00
$Public_t$	0.003**	0.05			0.002	0.18
$Affiliate_t$			0.008***	0.00	0.005*	0.06
$Public_t \times Affiliate_t$					0.009**	0.02
$FirmSize_t$	0.003***	0.00	0.002***	0.00	0.002***	0.01
Age_t	0.000***	0.00	0.000***	0.00	0.000***	0.00
$Leverage_t$	−0.001***	0.00	−0.001***	0.00	−0.001***	0.00
ROA_t	0.074***	0.00	0.073***	0.00	0.075***	0.00
$Advertising_t$	0.553***	0.00	0.557***	0.00	0.563***	0.00
$PreGivingD_t$	0.031***	0.00	0.031***	0.00	0.031***	0.00
$Big4Audit_t$	0.001	0.65	0.000	0.77	0.000	0.94
IMR	0.001	0.41	0.000	0.89	0.000	0.73
Year fixed effects	Yes		Yes		Yes	
Industry fixed effects	Yes		Yes		Yes	
Adjusted R^2	6.88%		6.90%		6.91%	
Panel B: OLS regression results using propensity-score matched sample ($N = 16{,}000$)						
Intercept	−0.021	0.74	−0.016	0.81	−0.015	0.82
$Public_t$	0.000	0.87			−0.001	0.65
$Affiliate_t$			0.012***	0.00	0.006	0.19
$Public_t \times Affiliate_t$					0.012**	0.05
$FirmSize_t$	0.008***	0.00	0.006***	0.00	0.007***	0.00
Age_t	0.000**	0.05	0.000**	0.02	0.000**	0.03
$Leverage_t$	−0.005***	0.00	−0.005***	0.00	−0.005***	0.00
ROA_t	0.090***	0.00	0.090***	0.00	0.089***	0.00
$Advertising_t$	0.696***	0.00	0.703***	0.00	0.700***	0.00
$PreGivingD_t$	0.023***	0.00	0.023***	0.00	0.023***	0.00
$Big4Audit_t$	0.000	0.83	−0.001	0.51	−0.002	0.46
Year fixed effects	Yes		Yes		Yes	
Industry fixed effects	Yes		Yes		Yes	
Adjusted R^2	9.06%		9.12%		9.13%	
Panel C: OLS regression results using coarsened-exact matched sample ($N = 39{,}054$)						
Intercept	0.126	0.00	0.127***	0.00	0.129***	0.00
$Public_t$	0.003*	0.09			0.002	0.30
$Affiliate_t$			0.012***	0.00	0.008**	0.02
$Public_t \times Affiliate_t$					0.014***	0.01
$FirmSize_t$	0.003***	0.00	0.002***	0.00	0.002***	0.00
Age_t	0.000***	0.00	0.000***	0.00	0.000***	0.00
$Leverage_t$	−0.006***	0.00	−0.006***	0.00	−0.006***	0.00
ROA_t	0.091***	0.00	0.090***	0.00	0.091***	0.00
$Advertising_t$	0.619***	0.00	0.628***	0.00	0.626***	0.00
$PreGivingD_t$	0.030***	0.00	0.030***	0.00	0.030***	0.00
$Big4Audit_t$	−0.002	0.17	−0.003*	0.09	−0.003**	0.04
Year fixed effects	Yes		Yes		Yes	
Industry fixed effects	Yes		Yes		Yes	

Table 3 continued

Dependent variable = $Giving_t$	Model (1)		Model (2)		Model (3)	
	Coeff.	*p* value	Coeff.	*p* value	Coeff.	*p* value
Adjusted R^2	7.02%		7.06%		7.08%	

This table presents the regression results based on 69,558 firm-year observations from 2000 to 2014. Panel A reports the OLS regression results for the Heckman second-stage model, Panel B reports the OLS regression results using a propensity-score matching, and Panel C reports the OLS regression results using a coarsened-exact matching. See Appendix 1 and Sect. 3.2 for variable definitions. All continuous variables are winsorized at the 1 and 99% level

***, **, and * denote statistical significance at the 1, 5, and 10% level, respectively

Next, in model (3), the positive coefficient on *Affiliate* (δ_2) indicates that, among private firms, business group-affiliated firms make more charitable contributions than non-affiliated firms. A similar pattern is observed for public firms: the value of $\delta_2 + \delta_3$ (0.014) is significantly positive at the 0.01 level. That is, regardless of the listing status, business group affiliates make more charitable contributions than non-affiliates.

In all the three models, the coefficients on control variables are consistent with prior studies. *ROA* is positively associated with corporate charitable contribution, suggesting that corporations with slack resources or more profits are likely to make a larger amount of charitable contributions (Waddock and Graves 1997). Also, *Leverage* is negatively associated with corporate giving, indicating that creditors effectively monitor managers' opportunistic behaviors (Brown et al. 2006). The positive coefficient on *Advertising* supports an argument that charitable giving is an alternative and complementary means of improving a firm's public image (Fry et al. 1982). Last, the coefficient on *IMR* (the inverse Mills ratio) is insignificant, suggesting that there is no significant self-selection bias concern in our regression models.

Panel B reports results of OLS regressions using the propensity-score matched sample.[14] We find that while the coefficient on *Public* (β_1) in model (1) is of the predicted sign, but insignificant, the coefficient on *Affiliate* (γ_1) in model (2) and the coefficient on the interaction term of *Public* and *Affiliate* (δ_3) in model (3) are positive and statistically significant, supporting H2 and H3. Note that the effect of the listing status on charitable contributions is still significant among business group affiliates: that is, in model (3), the sum of the coefficient on *Public* and the coefficient on the interaction of *Public* and *Affiliate* ($\delta_1 + \delta_3 = 0.011$) is significantly different from zero. The result suggests that public group-affiliated firms make more charitable contributions than private group-affiliated firms.

Panel C of Table 3 provides the OLS regression when employing the coarsened exact matching.[15] We find that the coefficient on *Public* (β_1) in model (1), the coefficient on *Affiliate* (γ_1) in model (2), and the coefficient on the interaction term of *Public* and *Affiliate* (δ_3) in model (3) are all positive and statistically significant, supporting our H1, H2, and H3.

To summarize, the results in Table 3 are consistent with the hypotheses that public firms make more corporate charitable contributions than private firms (H1) and business group-affiliated firms make more corporate charitable contributions than non-affiliated firms (H2). Further, we find that business groups strategically allocate a greater amount of corporate giving to their public affiliates (H3).

Additional Analyses

Control–Ownership Wedge and Corporate Giving

In the preceding section, we document that business groups strategically assign a larger amount of corporate giving to their public affiliates than private affiliates. However, it is still unclear whether this strategic allocation is linked to tunneling that benefits controlling shareholders at the expense of minority shareholders. If tunneling incentives cause the group-wide asymmetric giving strategy, corporate charitable giving should be associated with corporate ownership structure across the business group (Johnson et al. 2000).

Tunneling literature suggests that conflicts of interest between minority shareholders and controlling shareholders lead to the central agency problem (Kim and Yi 2006). Controlling shareholders who have voting rights in excess of their cash flow rights are likely to pursue their own

[14] We find that the mean values of most firm characteristics are similar between public and private firms, indicating an effective matching between the two groups (untabulated).

[15] The sample size of CEM differs from that of PSM because CEM allows for a 1:N matching. Further, unlike PSM that condenses various dimensions of the covariates into a single dimension (that is, a summary score), CEM considers every aspect of the covariates. Thus, CEM is less vulnerable to the so-called random-matching problem of PSM.

Table 4 Influence of control–ownership wedge on charitable contribution (for group-affiliated observations)

Dependent variable = $Giving_t$	Model (1)		Model (2)	
	Coeff.	*p* value	Coeff.	*p* value
(N = 4683)				
Intercept	−0.086***	0.00	−0.083***	0.00
$Wedge_t$	0.013**	0.05	0.007	0.35
$Wedge_t \times Public_t$			0.041**	0.03
$Public_t$			−0.010	0.30
$FirmSize_t$	0.016***	0.00	0.016***	0.00
Age_t	0.000**	0.02	0.000**	0.05
$Leverage_t$	−0.002***	0.01	−0.002***	0.01
ROA_t	0.180***	0.00	0.184***	0.00
$Advertising_t$	0.522***	0.00	0.504***	0.00
$PreGivingD_t$	0.033***	0.00	0.033***	0.00
$Big4Audit_t$	−0.007	0.14	−0.008	0.11
Year fixed effects	Yes		Yes	
Industry fixed effects	Yes		Yes	
Adjusted R^2	13.90%		13.98%	

This table presents the results from the OLS regression based on 4683 group affiliates. The dependent variable is corporate giving *(Giving)*. See Appendix 1 for variable definitions. All continuous variables are winsorized at the 1 and 99% level

***, **, and * denote statistical significance at the 1, 5, and 10% level, respectively

interests at the expense of minority shareholders. As the disparity between voting rights and cash flow rights (control–ownership wedge) widens, controlling shareholders have greater incentives to expropriate firm resources for private welfares bearing smaller costs of doing so (Joh 2003). Thus, we examine whether the strategic allocation of corporate giving is associated with control–ownership wedge in the business group setting.

Table 4 reports the estimation results of regressions of corporate charitable contributions on control–ownership wedge for large business group affiliates.[16] In model (1), we find that the coefficient on *Wedge* is positive and statistically significant, supporting an idea that business groups strategically use corporate charitable contributions as a channel of the tunneling process. In model (2), we find that the coefficient on *Wedge* is insignificant, but *Wedge x Public* is positive and statistically significant. That is, tunneling (greater charitable giving due to control–ownership disparity) is observed for public affiliates, but not for private affiliates. Alternatively, one may interpret the result such that between public and private firms with the same level of wedge, controlling shareholders of a business group would be more likely to do charity work through public affiliates rather than private affiliates. Even after controlling for the effect of wedge on corporate giving, we contend that, for business group charitable contributions, it would be more effective to utilize public affiliates than private affiliates because public firms are more visible than private firms.[17]

Public Affiliates' Charitable Contributions and Private Affiliates' Future Revenue Growth

In this subsection, we investigate potential benefits from business groups' strategic coordination of their affiliates' corporate giving. In particular, we examine whether the beneficial marketing effects, owing to public affiliates' charitable contributions, are transferred to private affiliates for the benefit of the controlling shareholders of business groups.

Panel A of Table 5 presents the regression results of private affiliates' future revenue growth ($\Delta SALE_t$). First, we find that the coefficient on $\Delta Giving_{t-1}$ is insignificant, suggesting that growth in the amount of private affiliates' charitable contributions does not affect their future revenue growth. However, we find in model (2) that the coefficient on $\Delta Group\text{−}wideGivingt_{t-1}$ is positive and statistically significant, suggesting that growth in the amount of business group-level charitable contributions is positively associated with private affiliates' future revenue growth. Lastly, in model (3), we find that the coefficient on

[16] Analyses are conducted for a sample of large business group affiliates over the period from 2001 to 2013 for which ownership structure data are available from KFTC.

[17] We are very grateful to an anonymous reviewer to point out this perspective.

Table 5 Influence of giving growth on sales growth within Business Group

Dependent variable = $\Delta SALE_t$	Model (1)		Model (2)		Model (3)	
	Coeff.	p value	Coeff.	p value	Coeff.	p value
Panel A: Private affiliates ($N = 1720$)						
Intercept	0.069***	0.00	0.069***	0.00	0.069***	0.00
$\Delta Giving_{t-1}$	−1.136	0.82	−1.792	0.72	−2.626	0.61
$\Delta Group\text{-}wideGiving_{t-1}$			7.073*	0.07		
$\Delta PublicGiving_{t-1}$					5.958**	0.04
$\Delta PrivateGiving_{t-1}$					5.963	0.34
$\Delta SALE_{t-1}$	0.213***	0.00	0.209***	0.00	0.208***	0.00
$\Delta SALE_{t-2}$	−0.008*	0.39	−0.009	0.37	−0.009	0.38
$\Delta Advertising_{t-1}$	−1.171	0.38	−1.116	0.40	−1.089	0.41
$\Delta Group\text{-}wideAdvertising_{t-1}$	0.392	0.84	0.052	0.98	−0.062	0.97
$FirmSize_t$	0.002	0.62	0.002	0.61	0.002	0.60
Adjusted R^2	4.38%		4.51%		4.57%	
Panel B: Public affiliates ($N = 1526$)						
Intercept	0.089***	0.01	0.089***	0.00	0.077**	0.02
$\Delta Giving_{t-1}$	−5.100	0.32	−3.983	0.46	−4.791	0.38
$\Delta Group\text{-}wideGiving_{t-1}$			−3.659	0.54		
$\Delta PublicGiving_{t-1}$					−0.079	0.99
$\Delta PrivateGiving_{t-1}$					−7.738	0.11
$\Delta SALE_{t-1}$	0.052**	0.04	0.051**	0.04	0.054**	0.03
$\Delta SALE_{t-2}$	−0.021	0.35	−0.022	0.34	−0.023	0.32
$\Delta Advertising_{t-1}$	0.556	0.62	0.541	0.63	0.527	0.64
$\Delta Group\text{-}wideAdvertising_{t-1}$	4.992**	0.03	5.180**	0.03	4.945**	0.04
$FirmSize_t$	−0.001	0.90	−0.001	0.88	0.001	0.83
Adjusted R^2	0.39%		0.36%		0.42%	

This table presents the results from the OLS regression based on 3246 group affiliates. The dependent variable is sales growth *(ΔSALE)*. See Appendix 1 for variable definitions. All continuous variables are winsorized at the 1 and 99% level

***, **, and * denote statistical significance at the 1, 5, and 10% level, respectively

$\Delta PublicGiving_{t-1}$ is positive and statistically significant, suggesting that growth in the amount of public affiliates' charitable contributions is positively associated with the future revenue growth of private affiliates within the same business group.[18]

Panel B of Table 5 presents the regression results of public affiliates' future revenue growth. Unlike the results for private affiliates, we find that the coefficient on $\Delta Group\text{-}wideGiving_{t-1}$ is insignificant, implying that public affiliates do not benefit from the group-wide corporate giving.

In sum, while private affiliates do not benefit from their own corporate giving, they enjoy the marketing effect from their public affiliates' corporate giving. These results in Table 5 support H3 that business groups strategically allocate a larger amount of corporate charitable contributions to public affiliates as a means of tunneling (i.e., to transfer the marketing effect) within business groups.

Changes in Charitable Contributions Around Going Public

We addressed potential endogeneity concerns about listing status by employing Heckman's (1979) two-stage procedure in our main analysis. However, it is still plausible that some other factors may influence the listing status of a firm and the corporate giving simultaneously. If listing status is an important factor for corporate giving, the level of corporate giving would change when companies go public. Since we are examining the change in the level of corporate giving of the same firm, we may better control for the inherent differences in corporate giving between public and private firms.

[18] In untabulated tests, we verify that our results remain qualitatively the same when using log change variables [e.g., $log(Giving_{t-1}/Giving_{t-2})$] instead of change variables [e.g., $\Delta Giving_{t-1}$] to measure various growth variables.

Figure 1 shows the trend of corporate giving around going public. Using 240 firms which went public during the period 2000–2014, we find that the mean values of *Giving* in years prior to firms' initial public offering increase until the year of public offering (from 0.067 in $t-3$ to 0.093 in t) and then gradually decrease after the public offering. It is notable that the levels of *Giving* remain elevated after the public offering compared with those in the pre-offering period. Further, Table 6 reports the t test results on the differences in the level of corporate giving between the pre- and the post-IPO periods. Consistent with Fig. 1, the level of corporate giving in the post-IPO period is significantly higher than that in the pre-IPO period. Overall, the results are consistent with H1 that the listing status is an important factor to influence the level of corporate giving.

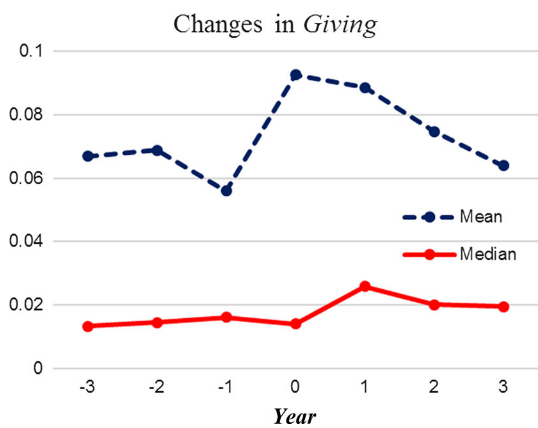

Fig. 1 Changes in corporate giving around going public. This figure illustrates yearly changes in the mean and median values of corporate charitable contributions (as a percentage of sales) around the IPO year of firms going public

Additional controls

Marketing assets

Prior literature (e.g., Wang and Qian 2011; Fry et al. 1982) argues that some firms use corporate giving as a means of corporate marketing activities. In that case, firms may use both corporate giving and conventional marketing activities like advertisement to enhance their public image and brand power. Suppose public (group-affiliated) firms are on average engaged with more marketing activities than private (non-affiliated) firms and use corporate charitable contributions for marketing purposes. If so, our results may not be attributable to the visibility and political costs as we hypothesized but rather attributable to the differential levels of marketing activities between public and private firms. To alleviate this concern, we additionally control for the potential effect from current and past marketing activities in Eqs. (2), (3), and (4).

Expenditures associated with current and past marketing activities are expensed rather than recorded as an asset in the financial statements. But, Lev and Sougiannis (1996) argue that advertising expenses on product promotion and brand development create an intangible asset, which we will refer to as a marketing asset. Accordingly, Lundholm and Sloan (2013) suggest that benefits from advertising need to be capitalized and amortized over an extended period. As in Lundholm and Sloan (2013), assuming that the effect of advertisement lasts about 3 years and

Table 6 Comparison of giving levels before and after going public

	Pre-IPO		Surrounding-IPO		Test of difference	
	Mean	Median	Mean	Median	t test	Wilcoxon Z test
Panel A: Pre-IPO (before, −1) versus surrounding-IPO (0, 1)						
Giving	0.066	0.013	0.091	0.020	0.00	0.00
N	(1110)		(465)			
	Pre-IPO		**Post-IPO**		**Test of difference**	
	Mean	**Median**	**Mean**	**Median**	**t test**	**Wilcoxon Z test**
Panel B: Pre-IPO (before, −1) versus Post-IPO (0, after)						
Giving	0.066	0.013	0.079	0.020	0.03	0.00
N	(1110)		(1074)			

All continuous variables are winsorized at the 1 and 99% level

***, **, and * denote statistical significance at the 1, 5, and 10% level, respectively

Table 7 Influence of listing status and group affiliation on charitable contribution with additional controls

Dependent variable = $Giving_t$	Model (1)		Model (2)		Model (3)	
	Coeff.	p value	Coeff.	p value	Coeff.	p value
Panel A: Control for marketing asset ($N = 69{,}558$)						
Intercept	0.066***	0.00	0.070***	0.00	0.070***	0.00
$Public_t$	0.003**	0.05			0.002	0.18
$Affiliate_t$			0.009***	0.00	0.005*	0.06
$Public_t \times Affiliate_t$					0.009**	0.02
$FirmSize_t$	0.003***	0.00	0.002***	0.00	0.002***	0.01
Age_t	0.000***	0.00	0.000***	0.00	0.000***	0.00
$Leverage_t$	−0.001***	0.00	−0.001***	0.00	−0.001***	0.00
ROA_t	0.074***	0.00	0.073***	0.00	0.074***	0.00
$Advertising_t$	0.485***	0.00	0.488***	0.00	0.487***	0.00
$PreGivingD_t$	0.031***	0.00	0.031***	0.00	0.031***	0.00
$Big4Audit_t$	0.001	0.65	0.000	0.78	0.000	0.94
$Unrecorded\text{-}Marketing\ Asset_t$	0.063	0.11	0.065*	0.10	0.066*	0.10
IMR	0.001	0.39	0.000	0.87	0.001	0.62
Year fixed effects	Yes		Yes		Yes	
Industry fixed effects	Yes		Yes		Yes	
Adjusted R^2	6.88%		6.90%		6.91%	
Panel B: Control for corporate governance ($N = 69{,}558$)						
Intercept	0.060***	0.00	0.069***	0.00	0.065***	0.00
$Public_t$	0.007***	0.00			0.006***	0.00
$Affiliate_t$			0.008***	0.00	0.005**	0.04
$Public_t \times Affiliate_t$					0.007*	0.09
$FirmSize_t$	0.002***	0.00	0.002***	0.01	0.001**	0.05
Age_t	0.000***	0.00	0.000***	0.00	0.000***	0.00
$Leverage_t$	−0.001***	0.00	0.000***	0.00	−0.001***	0.00
ROA_t	0.074***	0.00	0.073***	0.00	0.074***	0.00
$Advertising_t$	0.549***	0.00	0.556***	0.00	0.554***	0.00
$PreGivingD_t$	0.031***	0.00	0.031***	0.00	0.031***	0.00
$Big4Audit_t$	0.000	0.80	0.001	0.70	0.000	0.84
$CGVD_t$	0.009***	0.00	0.004**	0.08	0.008***	0.00
IMR	0.000	0.71	0.000	0.76	0.000	0.87
Year fixed effects	Yes		Yes		Yes	
Industry fixed effects	Yes		Yes		Yes	
Adjusted R^2	6.90%		6.90%		6.92%	

This table presents the results from the Heckman second-stage model after controlling for unrecorded-marketing asset and corporate governance. The dependent variable is corporate giving *(Giving)*. See Appendix 1 for variable definitions. All continuous variables are winsorized at the 1 and 99% level

***, **, and * denote statistical significance at the 1, 5, and 10% level, respectively

expenditures are made evenly throughout the year, we measure unrecorded marketing assets as the sum of 2.5/3 of current-year advertising expenditure, 1.5/3 of one-year-ago advertising expenditure, and 0.5/3 of two-year-ago advertising expenditure.[19]

Panel A of Table 7 presents the regression results for the influence of listing status and group affiliation on charitable contributions after controlling for unrecorded marketing assets. In all the models, we find qualitatively similar results with Panel A of Table 3, suggesting that our results are not driven by the marketing intensity of the firms.[20]

[19] In untabulated tests, we verify that our results, while become weak partly owing to the loss of data, remain qualitatively similar when we use a longer 10 years of capitalization period.

[20] In untabulated tests, we verify that our results remain qualitatively similar when we control for *Advertising Intensity* (advertising expense as a percentage of total sales) instead of *Unrecorded-Marketing Assets*.

Corporate Governance

While prior literature (e.g., Yermack 2009; Barnea and Rubin 2010) suggests that managers make use of corporate giving for their own interests, Masulis and Reza (2015) and Brown et al. (2006) show that good corporate governance can reduce agency problems like the opportunistic use of charitable contributions. The stock market (the government and regulators) may demand a higher level of corporate governance for public (group-affiliated) firms compared to private (non-affiliated) firms. If so, our results may not be attributable to the visibility and political costs of public firms as we hypothesized but rather attributable to the differential levels of corporate governance between public and private firms (between group-affiliated and non-affiliated firms). While we already control for the quality of external monitoring by including *Big4Audit* (an indicator variable that equals one if a firm has a Big 4 auditor) in Eqs. (2), (3), and (4), we address the issue further by controlling for the level of corporate governance in this subsection.

In particular, we suggest a proxy for corporate governance based on the Framework Act on Small and Medium Enterprises (SMEs).[21] In Korea, regulation on corporate governance are mainly aimed at larger firms and the government often offers SMEs exemptions from complying with the corporate law (e.g., the proportion of outside directors, see Black et al. 2006). Korea Corporate Governance Service (a non-profit organization jointly created by KRX, KLCA, and Korea Financial Investment Association) reports that SMEs have weaker corporate governance in various dimensions such as the board of directors, financial disclosures, audit committee, and corporate payouts compared to non-SMEs (Korea Corporate Governance Service 2012). Thus, we use whether a firm is designated as a SME under the Framework Act on SMEs to proxy for the level of a firm's corporate governance.

Panel B of Table 7 presents the regression results for the influence of listing status and group affiliation on charitable contributions after controlling for corporate governance. In all the models, we find qualitatively similar results with Panel A of Table 3, indicating that our results are not driven by the presence of monitors.

Political Corruption and Corporate Giving

Thus far, we implicitly assume that the motivation of corporate giving is to improve the firm's marketing and public relations. However, research suggests that some firms engage in corporate giving to improve their political connection or to manipulate the democratic system (e.g., Wang and Qian 2011; Li et al. 2015a, b). In this subsection, we separate out the effect of the former (i.e., social relations) from that of the latter (i.e., political corruption).[22] In particular, following OECD's foreign bribery report (2014), we categorize extractive, construction, and transportation & storage industries as political corruption-prone industries. We then rerun the main regression analysis separately for the two subgroups: a subgroup of firms in corruption-prone industries and a subgroup of firms in other industries. Table 8 provides the regression results. Panel A shows that when we confine our analysis to firms in the corruption-prone industries, none of our main hypotheses are supported. However, in Panel B, we find that all of our main hypotheses (H1, H2, and H3) continue to hold for other firms which would consider societal relations for their corporate giving decisions. These results suggest that our main findings are mostly attributable to those firms which are likely to be motivated to use charitable contributions for societal reasons (e.g., marketing and public relations), but firms in corruption-prone industries would use charitable contributions or other means such as direct bribes to influence government officials rather than to enhance corporate image or to maintain a good rapport with customers.

Alternative Definitions of Public Firms and Business Groups

Thus far, we define public firms as those that are listed on KRX, the nation's sole securities exchange, and business groups using group affiliation data on the KLCA database, which collects information regarding group affiliation from the financial statements. In this subsection, we extend the definitions of public firms and business groups.

First, KRX in fact has two separate equity markets: KOSPI and KOSDAQ. The former is often compared to the New York Stock Exchange (NYSE) and considered as the leading equity market for established publicly traded companies in the nation. The latter is often compared to NASDAQ and tends to list small technology-oriented growth firms. Thus, KOSDAQ listed firms would be more interested in making short-term profits from risky investments rather than pursuing a long-term value of the firms. Panel A of Table 9 shows that KOSPI listed firms make more charitable contributions than private firms while KOSDAQ listed firms do not, suggesting that listing on

21 Korean Corporate Governance Index (KCGI) developed by Black et al. (2006) is most widely used proxy for corporate governance in Korean evidence. However, KCGI is not feasible for our study, since it does not cover private firms.

22 We are very grateful to an anonymous reviewer to suggest isolating the impact of societal relations from that of political corruption.

Table 8 Political corruption and corporate giving

Dependent variable = $Giving_t$	Model (1)		Model (2)		Model (3)	
	Coeff.	*p* value	Coeff.	*p* value	Coeff.	*p* value
Panel A: Firms in political corruption-prone industries ($N = 9242$)						
Intercept	0.050***	0.00	0.049***	0.00	0.050***	0.00
$Public_t$	−0.005	0.37			−0.007	0.28
$Affiliate_t$			−0.020***	0.00	−0.027***	0.00
$Public_t \times Affiliate_t$					0.019	0.13
$FirmSize_t$	0.003*	0.06	0.004**	0.02	0.004**	0.02
Age_t	0.000	0.97	0.000	0.69	0.000	0.66
$Leverage_t$	−0.001	0.21	−0.001	0.18	−0.001	0.18
ROA_t	−0.007	0.71	−0.007	0.69	−0.007	0.69
$Advertising_t$	0.554**	0.02	0.546**	0.03	0.542**	0.03
$PreGivingD_t$	0.029***	0.00	0.029***	0.00	0.029***	0.00
$Big4Audit_t$	0.004	0.39	0.006	0.19	0.007	0.14
IMR	−0.001	0.72	−0.001	0.71	−0.001	0.65
Year fixed effects	Yes		Yes		Yes	
Industry fixed effects	Yes		Yes		Yes	
Adjusted R^2	2.79%		2.88%		2.88%	
Panel B: Firms in industries not prone to political corruption ($N = 60{,}316$)						
Intercept	0.068***	0.00	0.073***	0.00	0.072***	0.00
$Public_t$	0.004**	0.02			0.003**	0.04
$Affiliate_t$			0.012***	0.00	0.009***	0.00
$Public_t \times Affiliate_t$					0.009**	0.03
$FirmSize_t$	0.002***	0.00	0.001**	0.05	0.001	0.14
Age_t	0.000***	0.00	0.000***	0.00	0.000***	0.00
$Leverage_t$	−0.001***	0.00	−0.001***	0.00	−0.001***	0.00
ROA_t	0.085***	0.00	0.084***	0.00	0.086***	0.00
$Advertising_t$	0.552**	0.00	0.558***	0.00	0.559***	0.00
$PreGivingD_t$	0.031***	0.00	0.031***	0.00	0.031***	0.00
$Big4Audit_t$	0.000	0.87	0.000	0.96	−0.001	0.60
IMR	0.001	0.46	0.000	0.90	0.000	0.72
Year fixed effects	Yes		Yes		Yes	
Industry fixed effects	Yes		Yes		Yes	
Adjusted R^2	7.62%		7.66%		7.67%	

This table presents the OLS regression results for the Heckman second-stage model for sub-samples: Industries prone to political corruption (*Panel A*) and industries not prone to political corruption (*Panel B*). See Appendix 1 and Sect. 5.5 for variable definitions. All continuous variables are winsorized at the 1 and 99% level

***, **, and * denote statistical significance at the 1, 5, and 10% level, respectively

KOSPI only influences corporate philanthropic decisions.[23]

Next, we examine whether larger business groups, also known as *Chaebols*, exhibit a similar pattern of philanthropic behavior as smaller business groups. Prior literature highlights that large business groups in Korea or Chaebols (Chang 2003; Joh 2003) have significant economic importance for the Korean economy (Bae et al. 2008; Baek et al. 2006). Chang and Hong (2000) argue that Chaebols play a substantial role in determining the economic performance of Korea. Kim and Yi (2006) report that Chaebols manage earnings more aggressively than non-affiliated firms. Given that Chaebols exhibit different patterns of corporate decisions from small business groups, they may exhibit different patterns of corporate giving behavior. Thus, we examine whether our findings are robust to

[23] We are very grateful to an anonymous reviewer for pointing out this market segregation in Korea.

Table 9 Alternative definitions of public firms and business groups

Dependent variable = $Giving_t$	Coeff.	p value
Panel A: Distinction between KOSPI & KOSDAQ listing (N = 69,558)		
Intercept	0.072***	0.00
$KOSPI_t$	0.010***	0.00
$KOSDAQ_t$	−0.001	0.39
$FirmSize_t$	0.002***	0.00
Age_t	0.000***	0.00
$Leverage_t$	−0.001***	0.00
ROA_t	0.073***	0.00
$Advertising_t$	0.552***	0.00
$PreGivingD_t$	0.031***	0.00
$Big4Audit_t$	0.000	0.78
Year fixed effects	Yes	
Industry fixed effects	Yes	
Adjusted R^2	6.91%	

Dependent variable = $Giving_t$	Model (1)		Model (2)		Model (3)	
	Coeff.	p value	Coeff.	p value	Coeff.	p value
Panel B: KFTC designated large business groups or Chaebols (*G_KFTC*) (N = 69,558)						
Intercept	0.066***	0.00	0.070***	0.00	0.070***	0.00
$Public_t$	0.003**	0.05			0.002	0.25
G_KFTC_t			0.007***	0.00	0.001	0.58
$Public_t \times G\_KFTC_t$					0.013***	0.00
$FirmSize_t$	0.003***	0.00	0.002***	0.00	0.002***	0.01
Age_t	0.000***	0.00	0.000***	0.00	0.000***	0.00
$Leverage_t$	−0.001***	0.00	−0.001***	0.00	−0.001***	0.00
ROA_t	0.074***	0.00	0.073***	0.00	0.074***	0.00
$Advertising_t$	0.553***	0.00	0.556***	0.00	0.556***	0.00
$PreGivingD_t$	0.031***	0.00	0.031***	0.00	0.031***	0.00
$Big4Audit_t$	0.001	0.65	0.001	0.70	0.000	0.96
IMR	0.001	0.41	0.000	0.88	0.000	0.70
Year fixed effects	Yes		Yes		Yes	
Industry fixed effects	Yes		Yes		Yes	
Adjusted R^2	6.88%		6.89%		6.91%	

Dependent variable = $Giving_t$	Model (1)		Model (2)		Model (3)	
	Coeff.	p value	Coeff.	p value	Coeff.	p value
Panel C: Top 30 large business groups (*Big30*) (N = 69,558)						
Intercept	0.066***	0.00	0.070***	0.00	0.071***	0.00
$Public_t$	0.003**	0.05			0.001	0.38
$Big30_t$			0.009***	0.00	0.000	0.99
$Public_t \times Big30_t$					0.022***	0.00
$FirmSize_t$	0.003***	0.00	0.002***	0.00	0.002***	0.01
Age_t	0.000***	0.00	0.000***	0.00	0.000***	0.00
$Leverage_t$	−0.001***	0.00	−0.001***	0.00	−0.001***	0.00
ROA_t	0.074***	0.00	0.073***	0.00	0.074***	0.00
$Advertising_t$	0.553***	0.00	0.556***	0.00	0.557***	0.00
$PreGivingD_t$	0.031***	0.00	0.031***	0.00	0.031***	0.00
$Big4Audit_t$	0.001	0.65	0.001	0.70	0.000	0.96

Table 9 continued

Dependent variable = $Giving_t$	Model (1)		Model (2)		Model (3)	
	Coeff.	p value	Coeff.	p value	Coeff.	p value
IMR	0.001	0.41	0.000	0.90	0.000	0.78
Year fixed effects	Yes		Yes		Yes	
Industry fixed effects	Yes		Yes		Yes	
Adjusted R^2	6.88%		6.90%		6.94%	

Panel A presents the results from the OLS regression based on 69,558 firm-year observations. The dependent variable is corporate giving (*Giving*). See Appendix 1 for variable definitions. All continuous variables are winsorized at the 1 and 99% level. Panels B and C present the results from OLS regression for the Heckman second-stage model based on 69,558 firm-year observations from 2000 to 2014. The dependent variable is corporate giving *(Giving)*. In order to examine the influence of large business groups (i.e., Chaebols), we replace our variable of interest, *Affiliate*¸ with *G_KFTC* or *Big30*. Panel B reports the results based on *G_KFTC*, which represents large business groups designated by KFTC and Panel C reports the results based on *Big30*, which represents one of the 30 largest business groups in Korea. See Appendix 1 and Sect. 5.5 for variable definitions. All continuous variables are winsorized at the 1 and 99% level

***, **, and * denote statistical significance at the 1, 5, and 10% level, respectively

different definitions of business groups based on the information disclosed from KFTC.[24]

Panel B of Table 9 reports the regression results of Eqs. (2), (3), and (4) by replacing *Affiliate* with *G_KFTC*, which represents large business groups or Chaebols as identified by KFTC.[25] We find significant and positive coefficients on *G_KFTC* in model (2), and on *Public* x *G_KFTC* in model (3), suggesting that Chaebols make more charitable contributions than non-Chaebols and that they also strategically allocate corporate giving to public affiliates. Panel C of Table 9 presents qualitatively similar results using the largest thirty Chaebols (*Big30)*. Furthermore, we find out that the coefficient on *Public* × *Big30* (0.022) is significantly larger than that on *Public* × *Affiliate* (0.009) in Panel A of Table 3, implying that the strategic allocation of corporate giving is more pronounced among large Chaebols than among smaller business groups in Korea.

Corporate Giving Decision

In this subsection, we investigate whether the influence of listing status and group affiliation on the likelihood of corporate giving exhibits a different pattern from that on the amount of corporate giving. Using *GivingD*, a dummy variable that equals one if a firm ever makes charitable contributions, as the dependent variable, we estimate the following Probit model:

$$\begin{aligned} Pr\left(GivingD_{i,t} = 1\right) = \Phi(&\delta_0 + \delta_1 Public_{i,t} + \delta_2 Affiliate_{i,t} \\ &+ \delta_3 Public_{i,t} \times Affiliate_{i,t} \\ &+ \delta_4 FirmSize_{i,t} + \delta_5 Age_{i,t} + \delta_6 Leverage_{i,t} + \delta_7 ROA_{i,t} \\ &+ \delta_8 Advertising_{i,t} + \delta_9 PreGivingD_{i,t} + \delta_{10} Big4Audit_{i,t} \\ &+ (\text{Year dummies}) + (\text{Industry dummies}) + \varepsilon) \end{aligned} \quad (5)$$

Panel A of Table 10 reports the Probit regression results of Eq. (5). We find a positive coefficient on *Public*, suggesting that public firms are more likely to make charitable contributions than private firms. However, unlike the results of giving amount, the coefficients on *Affiliate* and *Public* × *Affiliate* are not statistically significant, implying that business group-affiliated firms are not more likely to make charitable contributions than non-affiliated firms. Overall, the above results suggest that while higher political costs and visibility costs affect business group-affiliated firms to increase the amount of charitable contributions, the costs do not appear to influence the corporate giving decision of whether those affiliated firms ever make charitable contributions.[26]

Finally, we examine the initiation effect of corporate giving on sales growth as it could be an important event in corporate philanthropy policy. We regress sales growth on an indicator variable for the initiation of corporate giving (*Initiation*) as follows:

$$\begin{aligned} \Delta SALE_{i,t+1} = &\ \delta_0 + \delta_1 \Delta Giving_{i,t} + \delta_2 Initiation_{i,t} \\ &+ \delta_3 \Delta Giving_{i,t} \times Initiation_{i,t} \\ &+ \delta_4 \Delta SALE_{i,t} + \delta_5 \Delta SALE_{i,t-1} + \delta_6 \Delta Advertising_{i,t} \\ &+ \delta_7 FirmSize_{i,t} + \varepsilon \end{aligned} \quad (6)$$

[24] Bae et al. (2008) and Kim and Yi (2006) highlight the economic importance of thirty largest Chaebols in the Korean economy.

[25] Defining a Chaebol as "a group of companies of which more than 30% of shares are owned by the group's controlling shareholder and its affiliated companies," KFTC regularly releases the list of Chaebols and ranks the business groups by their total assets.

[26] We find the same results from Logit regression estimates (untabulated).

Table 10 corporate giving decision

Dependent variable = $GivingD_t$	Model (1)		Model (2)		Model (3)	
	Coeff.	p value	Coeff.	p value	Coeff.	p value
Panel A: Influence of listing status and group affiliation on the likelihood of corporate giving ($N = 111{,}525$)						
Intercept	−1.265***	0.00	−1.289***	0.00	−1.267***	0.00
$Public_t$	0.071***	0.00			0.071***	0.00
$Affiliate_t$			−0.016	0.52	−0.011	0.71
$Public_t \times Affiliate_t$					−0.004	0.94
$FirmSize_t$	0.174***	0.00	0.183***	0.00	0.175***	0.00
Age_t	−0.007***	0.00	−0.008***	0.00	−0.007***	0.00
$Leverage_t$	0.005***	0.00	0.005***	0.00	0.005***	0.00
ROA_t	1.145***	0.00	1.118***	0.00	1.145***	0.00
$Advertising_t$	4.547***	0.00	4.541***	0.00	4.544***	0.00
$PreGivingD_t$	1.701***	0.00	1.702***	0.00	1.701***	0.00
$Big4Audit_t$	−0.013	0.34	−0.003	0.84	−0.012	0.38
Year fixed effects	Yes		Yes		Yes	
Industry fixed effects	Yes		Yes		Yes	
Adjusted R^2	50.40%		50.39%		50.40%	

Dependent variable = $\Delta SALE_{t+1}$	Coeff.	p value
Panel B: Influence of the initiation of corporate giving on sales growth ($N = 55{,}341$)		
Intercept	−0.002	0.57
$\Delta Giving_t$	−0.594	0.41
$Initiation_t$	0.022***	0.01
$\Delta Giving_t$ x $Initiation_t$	14.192***	0.00
$\Delta SALE_t$	0.008*	0.07
$\Delta SALE_{t-1}$	−0.053***	0.00
$\Delta Advertising_t$	1.683***	0.00
$FirmSize_t$	0.019***	0.00
Adjusted R^2	0.98%	

Panel A presents the results from Probit regressions based on 111,525 firm-year observations. The dependent variable is an indicator variable for corporate giving decision (*GivingD)*. See Appendix 1 for variable definitions. All continuous variables are winsorized at the 1 and 99% level. Panel B presents the results from OLS regressions based on 55,341 firm-year observations. The dependent variable is sales growth *(ΔSALE)*. See Appendix 1 for variable definitions. All continuous variables are winsorized at the 1 and 99% level

***, **, and * denote statistical significance at the 1, 5, and 10% level, respectively

In Panel B of Table 10, we find that the coefficient on *Initiation* is significantly positive at the one percent level, suggesting that the initiation of corporate giving is significantly associated with subsequent revenue growth. Further, the coefficient on the interaction of *Initiation* and giving growth is also significantly positive at the one percent level, that is, the beneficial effect of increasing corporate giving on subsequent revenue growth is conspicuous when firms initiate their giving.[27]

[27] We are very grateful to an anonymous reviewer for the suggestion to examine the initiation effect of corporate giving.

Conclusion

We examine whether firm-specific corporate governance characteristics, in particular listing status and business group affiliation, are associated with corporate charitable contributions. We find that public firms make more charitable contributions than private firms and that business group-affiliated firms make more corporate contributions than non-affiliated firms. Further, we find that greater charitable contributions by public firms vis-à-vis private firms are more pronounced for business group-affiliated firms than for non-affiliated firms. Furthermore, our results are robust to alternative definitions of business groups (i.e., top 30 Chaebols). Overall, our results suggest that public firms which are under greater public scrutiny and business groups which are subject to higher political costs and visibility costs are more likely to engage in corporate social activities and also that business groups strategically make use of corporate philanthropy as a means of tunneling corporate resources to controlling shareholders.

This study is subject to several caveats. We predict and provide empirical evidence consistent with public firms making more corporate giving due to visibility and political costs (e.g., Merton 1987) and business groups utilizing corporate giving by public affiliates as a means of tunneling (e.g., Johnson et al. 2000). We have been careful in addressing potential endogeneity between listing status and corporate giving by employing several matching techniques in the literature including the Heckman two-stage procedure. However, to the extent that these procedures are not effective, one should be cautious in drawing causal inferences. Accordingly, we do not exclude the possibility that business groups and their public affiliates might simply have a greater need to make a large amount of corporate giving as good corporate citizens and as part of their strategic investments to look good to stakeholders including customers thereby enhancing their long-term firm value (e.g., Lev et al. 2010). Future research may extend the study by examining the long-term consequences of a large amount of corporate giving with respect to business group affiliates' value and performance.

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Appendix 1: Variable Descriptions

Variable	Definition
Giving	The amount of charitable contributions as a percentage of sales
GivingD	An indicator variable that equals one if a firm makes charitable contributions
Public	An indicator variable that equals one if a firm is listed on KRX
Affiliate	An indicator variable that equals one if a firm belongs to a business group
FirmSize	The natural log of total assets
Age	The number of years elapsed since the incorporation
Leverage	The book value of short- and long-term debt scaled by the book value of equity
ROA	Net income divided by total assets
Advertising	Advertising expense scaled by total assets
PreGivingD	An indicator variable that equals one if a firm made charitable contributions in the preceding year
Big4Audit	An indicator variable that equals one if a firm has a big 4 auditor
ExportRatio	The ratio of the amount of exports to total sales
QuickRatio	The quick ratio measured by current assets less inventory divided by current liabilities
Industry_PubPr	Proportion of public firms in a firm's industry
HHI	Herfindahl–Hirschman index, measured as the sum of squares of the market shares (firm sales/industry sales) of the firms in the industry
Δ*SALE*	Percentage change in sales
Δ*Giving*	Change in the charitable contributions, scaled by sales
Δ*PublicGiving*	Change in the sum of the charitable contributions of public affiliates, scaled by the sum of the sales of public affiliates
Δ*PrivateGiving*	Change in the sum of the charitable contributions of private affiliates, scaled by the sum of the sales of private affiliates
Δ*Group-wide Giving*	Change in the sum of the charitable contributions of all the group affiliates, scaled by the sum of the sales of group affiliates
Δ*Group-wide Advertising*	Change in the sum of the advertising expense of all the group affiliate, scaled by the sum of the sales of group affiliates
Wedge	The difference between control rights and cash flow rights held by the controlling shareholder
Unrecorded-Marketing Asset	$(2.5/3) \times Advertising_t + (1.5/3) \times Advertising_{t-1} + (0.5/3) \times Advertising_{t-2}$
CGVD	An indicator variable that equals one if a firm is not classified as a small and medium size enterprise according to *Framework Act on Small and Medium Enterprises*
G_KFTC	An indicator variable for large business groups or Chaebols as designated by KFTC
KOSPI	An indicator variable for firms on the KOSPI market
KOSDAQ	An indicator variable for firms on the KOSDAQ market
Big30	An indicator variable that equals one if a firm belongs to top 30 Chaebol
Initiation	An indicator variable for the initiation of corporate giving by firms that have not made charitable contributions in the preceding three years

Appendix 2: Description on the Statistical Model

$$\begin{aligned} Giving_{i,t} = {} & \delta_0 + \delta_1 Public_{i,t} + \delta_2 Affiliate_{i,t} + \delta_3 Public_{i,t} \\ & \times Affiliate_{i,t} + \delta_4 FirmSize_{i,t} + \delta_5 Age_{i,t} \\ & + \delta_6 Leverage_{i,t} + \delta_7 ROA_{i,t} + \delta_8 Advertising_{i,t} \\ & + \delta_9 PreGivingD_{i,t} + \delta_{10} Big4Audit_{i,t} + \delta_{11} IMR \\ & + (\text{Year dummies}) + (\text{Industry dummies}) + \varepsilon, \end{aligned} \quad (4)$$

	Public	Private
Business Group	*Public* × *Affiliate* [A]	*Private* × *Affiliate* [C]
Non-Group	*Public* × *Non-affiliate* [B]	*Private* × *Non-affiliate* [D]

A: $\delta_0 + \delta_1 + \delta_2 + \delta_3$, B: $\delta_0 + \delta_1$, C: $\delta_0 + \delta_2$, D:δ_0, $A - B = \delta_2 + \delta_3$, $A - C = \delta_1 + \delta_3$, $B - D = \delta_1$, $C - D = \delta_2$

Equation (4) includes two variables of interest, *Public* and *Affiliate,* and an interaction variable of the two. In effect, the sample is partitioned into four subgroups: Public and affiliated firms [A], Public, but not affiliated firms [B], Private but affiliated [C], and Private and not affiliated firms [D]. δ_1 is an estimate of the effect of listing status on corporate charitable contributions and δ_2 is an estimate of the effect of group affiliation on corporate charitable contributions. δ_3 is an estimate of the excessive effect of subgroup A on corporate charitable contributions. The impact of subgroup A (B) on charitable contributions, on average, is computed by summing up δ_0, δ_1, δ_2, and δ_3 (δ_0 and δ_1), while the impact of subgroup C (D) on charitable contributions is sum of δ_0 and δ_2 (δ_0). The different impact on charitable contributions between A and B is the sum of δ_2 and δ_3, while it between A and C is the sum of δ_1 and δ_3. δ_1 (δ_2) shows the different influence on charitable contributions between B and D (C and D).

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